



RATIONAL FUNCTIONS

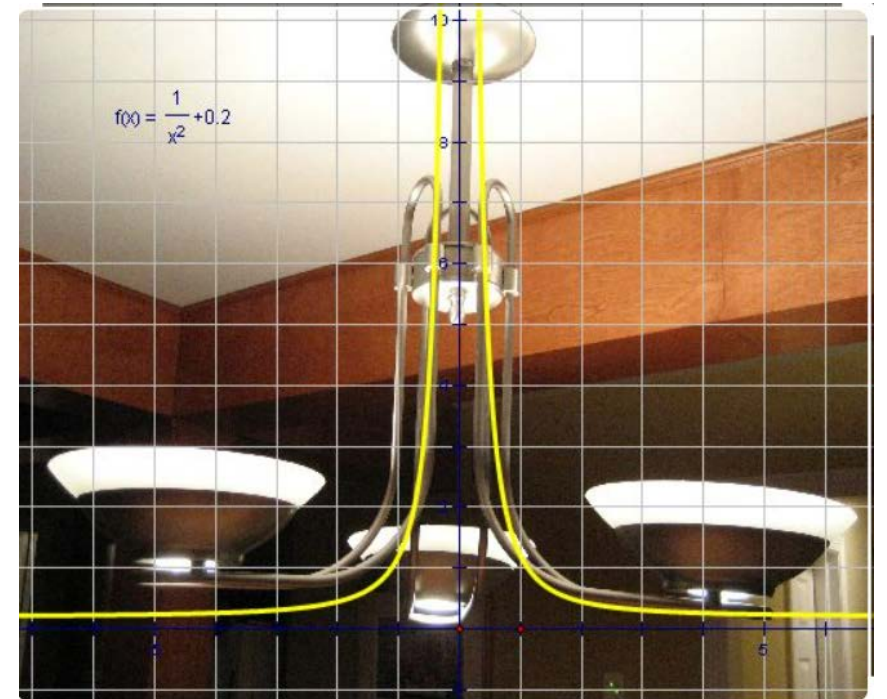
ASYMPTOTES AND GRAPHS



RATIONAL FUNCTION

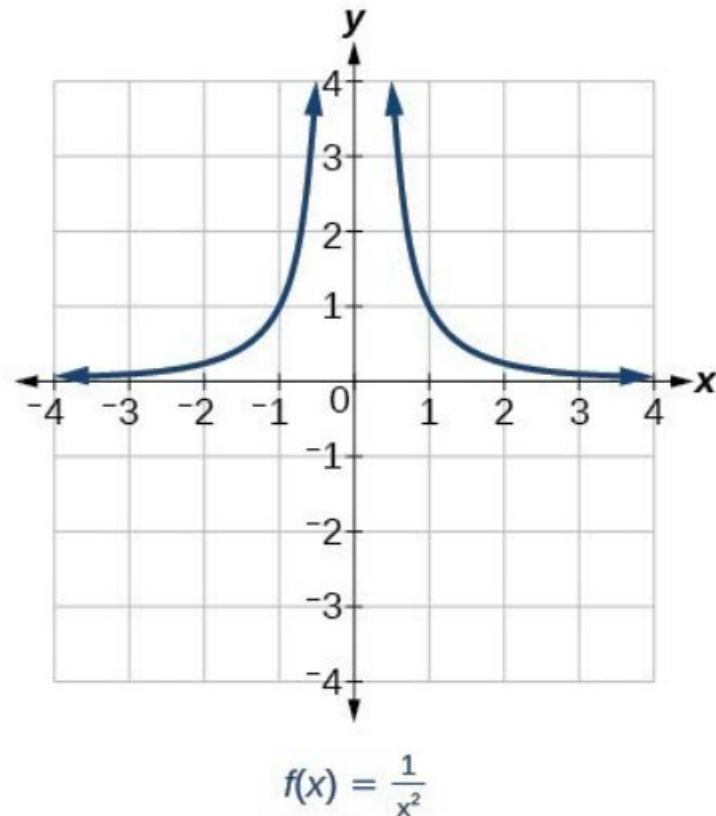
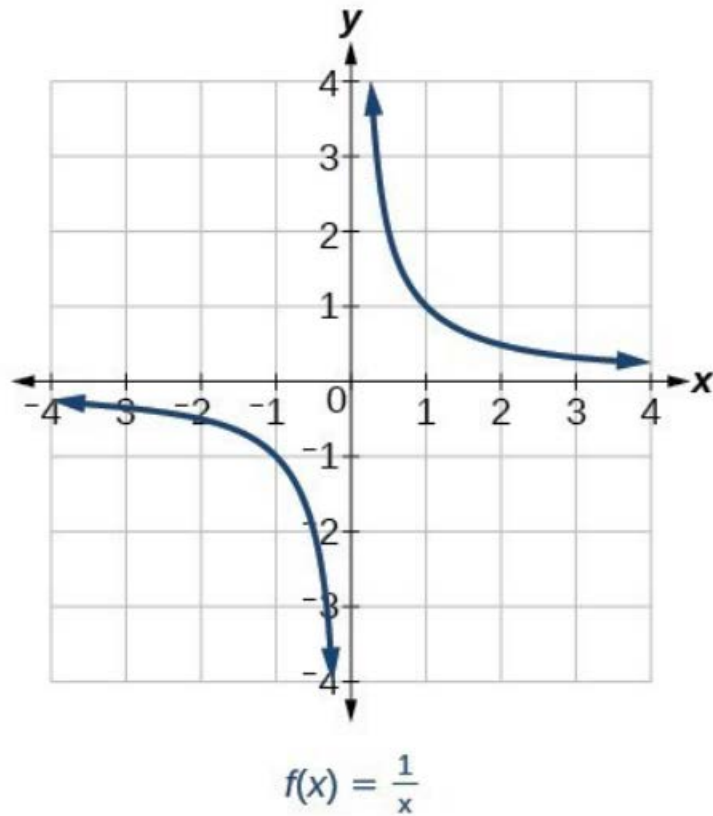
A **rational function** is a function that can be written as the quotient of two polynomial functions $P(x)$ and $Q(x)$.

$$f(x) = \frac{P(x)}{Q(x)} = \frac{a_px^p + a_{p-1}x^{p-1} + \dots + a_1x + a_0}{b_qx^q + b_{q-1}x^{q-1} + \dots + b_1x + b_0}, Q(x) \neq 0$$



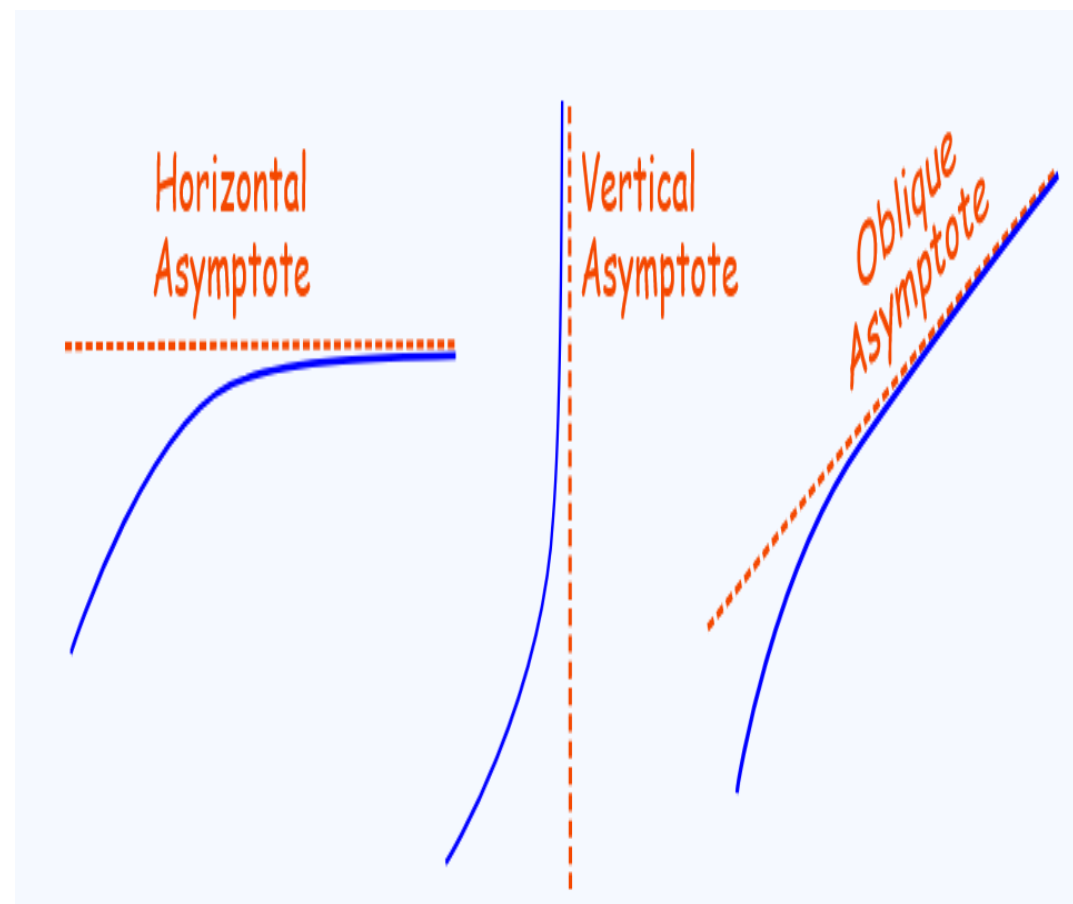
Since division by 0 is not allowed, we must exclude those values of x for which $q(x) = 0$.

$f(x) = \frac{1}{x}$, $f(x) = \frac{1}{x^2}$, $f(x) = \frac{x}{x^2-2}$ and $f(x) = -\frac{1}{x}$ are couple of examples of rational functions.



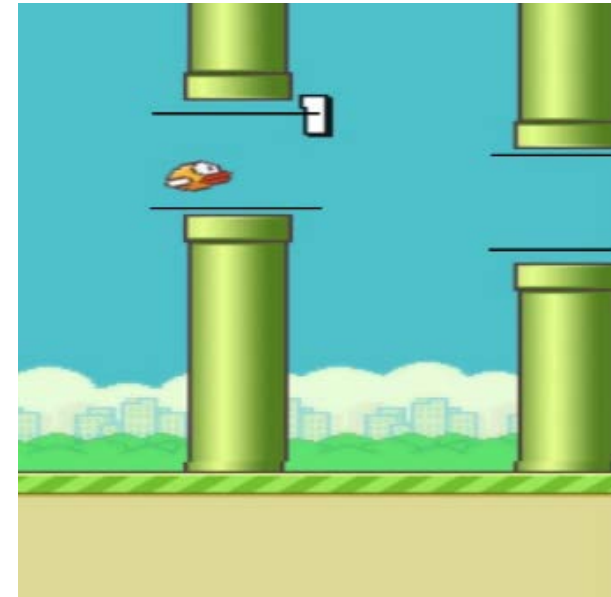
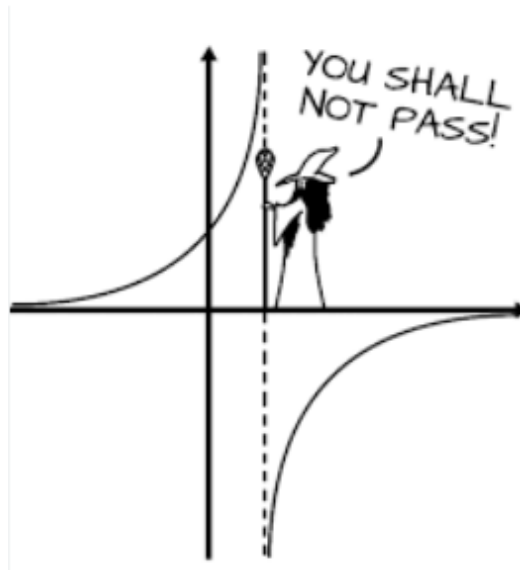
ASYMPTOTES

An **asymptote** of a curve is a line such that the distance between the curve and the line approaches zero as one or both of the x or y coordinates tends to infinity.



ASYMPTOTES

- There are two types of Asymptotes:
- Vertical asymptotes
- Horizontal asymptotes



VERTICAL ASYMPTOTES

The line $x = x_o$ is a vertical asymptote of the graph of $y = f(x)$ if the values of $f(x)$ approach $-\infty$ or ∞ as x approaches x_o (from the left or the right).

A vertical asymptote will occur on the graph of a function at an x -value at which the denominator (but not the numerator) of the function is equal to zero.

Vertical asymptotes of the Rational Function

The graph of the rational function $h(x) = \frac{f(x)}{g(x)}$ has a vertical asymptote at $x = c$, if $g(c) = 0$ and $f(c) \neq 0$.

HORIZONTAL ASYMPTOTES

A horizontal asymptote tells us the behavior of the values of the function (y -coordinates) when x increases or decreases without bound, we use limits at infinity to determine the existence of horizontal asymptotes.

Horizontal Asymptotes of Rational Functions

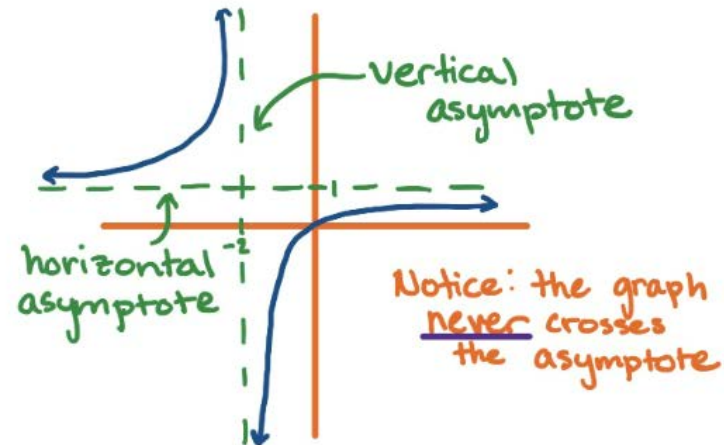
Consider the rational function

$$y = \frac{f(x)}{g(x)} = \frac{a_n x^n + \dots + a_1 x + a_o}{b_m x^m + \dots + b_1 x + b_o}$$

List	Case. 1	Case. 2	Case. 3
Condition	$n < m$	$n = m$	$n > m$
Horizontal Asymptote	$y = 0$	$y = \frac{a_n}{b_m}$	no horizontal asymptote

1. If $n < m$ (that is, if the degree of the numerator is less than that of the denominator), a horizontal asymptote occurs at $y = 0$ (the x-axis).
2. If $n = m$ (that is, if the degree of the numerator equals that of the denominator), a horizontal asymptote occurs at $y = \frac{a_n}{b_m}$ (the ratio of the leading coefficients).
3. If $n > m$ (that is, if the degree of the numerator is greater than that of the denominator), there is no horizontal asymptote.

Asymptotes



EXAMPLE

(a) Use values of x from -5 to 5 to develop a table of function values for the graph of

$$y = \frac{12x + 8}{3x - 9}$$

(b) Sketch the graph.

Solution:

(a) Because $3x - 9 = 0$ when $x = 3$, it follows that $x = 3$ is not in the domain of this function. The values in the table indicate that the graph is approaching a vertical asymptote at $x = 3$.

x	-5	-4	-3	-2	-1	0
y	2.17	1.90	1.56	1.07	0.33	-0.89
(x, y)	(-5, 2.17)	(-4, 1.90)	(-3, 1.56)	(-2, 1.07)	(-1, 0.33)	(0, -0.89)

1	2	2.5	2.9	3.1	3.5
-3.33	-10.7	-25.33	-142.67	150.67	33.33
(1,-3.33)	(2, -10.7)	(2.5, -25.33)	(2.9, -142.67)	(3.1, 150.67)	(3.5, 33.33)

4	5
18.67	11.33
(4, 18.67)	(5, 11.33)

(b) Using the information about vertical and horizontal asymptotes and plotting these points give the graph in Figure 2.

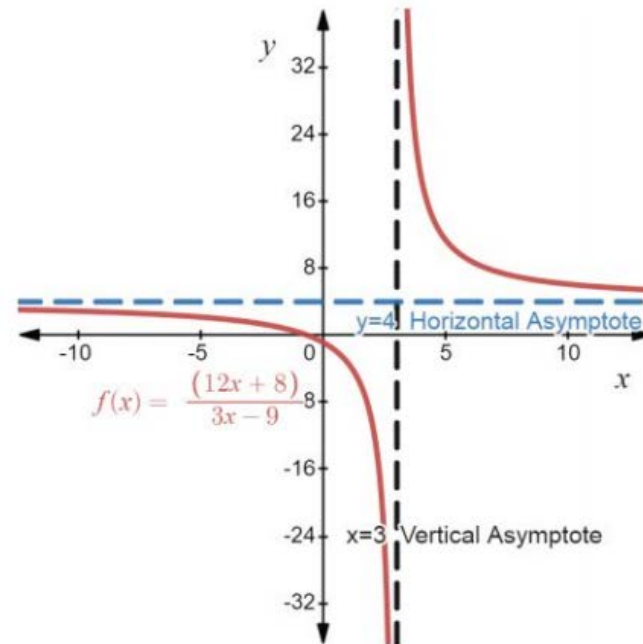


Figure. 2

EXAMPLE 2

The function $f(x) = \frac{1}{x}$ is a special function called a reciprocal function. A table of sample values and the graph of $y = \frac{1}{x}$ are shown in Figure. 3.

Because division by 0 is not possible, $x = 0$ is not in the domain of the function.─

x	y
-3	-0.33
-2	-0.5
-1	-1
-0.1	-10
-0.04	-25
-0.001	-1000
3	0.33
2	0.5
1	1
0.1	10
0.04	25
0.001	1000

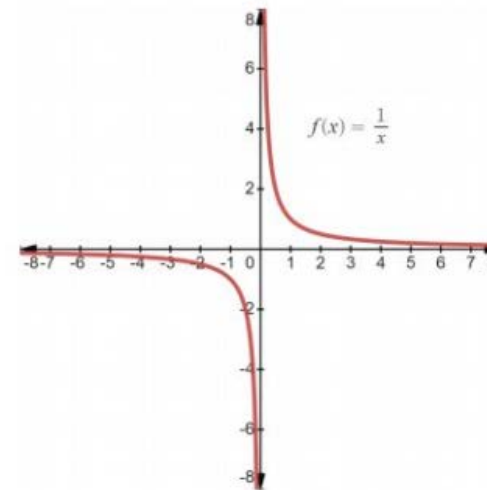


Figure. 3

EXERCISE

$$25. y = \frac{1 - 2x}{x}$$

$$26. y = \frac{x}{x - 1}$$

For the following exercises, make tables to show the behavior of the function near the vertical asymptote and reflecting the horizontal asymptote.

$$65. f(x) = \frac{1}{x - 2}$$

$$66. f(x) = \frac{x}{x - 3}$$

$$67. f(x) = \frac{2x}{x + 4}$$

$$68. f(x) = \frac{2x}{(x - 3)^2}$$

$$69. f(x) = \frac{x^2}{x^2 + 2x + 1}$$

82. The concentration C of a drug in a patient's bloodstream t hours after injection is given by $C(t) = \frac{2t}{3 + t^2}$. What happens to the concentration of the drug as t increases?

83. The concentration C of a drug in a patient's bloodstream t hours after injection is given by $C(t) = \frac{100t}{2t^2 + 75}$. Use a calculator to approximate the time when the concentration is highest.